MGMTMSA-400-01/02: Homework 1

Due Date: October 11, 2026, 11:59 PM
No Late Homework!

Submission: Submit write-up on Gradescope, code or workbooks on Bruin Learn

Note: This assignment has a total score of 40 points. Your score will be converted to 10 points toward the final grade.

Bonus Points: Question 3(b-4) is optional and counts only for bonus credit. The rest of the homework, i.e., Questions 1–3(b-3), adds up to 40 points in total. You can earn bonus points by answering 3(b-4) correctly, but the maximum score for this homework is still capped at 40. For example, if you earn 39 points from Questions 1–3(b-3) and 2 bonus points from 3(b-4), your final score will be 40 (not 41).

Question 1: Law of Large Numbers and Central Limit Theorem (14 points)

In this question, you will use simulation to explore the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT). Draw observations *independently* from the uniform distribution $\mathcal{U}[-5, 5]$. The population mean μ is the mean of this uniform distribution, which is zero.

Note: Run `set.seed(123)` once at the beginning, before generating any random numbers. This makes your results reproducible each time you run the document. You do not need to reset the seed before each part.

(a) Generate one sample for each sample size $N = 10$, 100, 1000, and 10000. For each sample, report the sample mean $\bar{Y}$ and its absolute difference from the population mean, $|\bar{Y} - \mu|$, in a table. (2 points)

(b) As sample size N increases, would you generally expect the absolute difference $|\bar{Y} - \mu|$ to become smaller or larger? Which theorem explains this expectation? Discuss using the results from (a). (2 points)

(c) Generate 10000 samples, each with $N = 10$ observations, and compute the mean of each sample. Plot a histogram of the resulting 10000 sample means. (2 points)

(d) Which theorem explains why the sample means in (c) have an approximately normal distribution, even though the original observations are uniformly distributed? (2 points)

(e) Repeat the simulation in (c) using sample sizes $N = 100$ and $N = 1000$, generating 10000 samples for each sample size. For each of the three sample sizes ($N = 10$, 100, and 1000), compute the 25th and 90th percentiles of the 10000 sample means. Report your results in a table. You may reuse your results from (c) for $N = 10$. (3 points)

In R, you can compute both percentiles of a vector `data` using `quantile(data, c(0.25, 0.90))`.

(f) As N increases, do the 25th and 90th percentiles in (e) move closer to or farther from the population mean of zero? Explain intuitively why increasing N leads to this pattern. (3 points)

Question 2: Jane's All-You-Can-Eat Restaurant (8 points)

Jane runs an all-you-can-eat restaurant that charges $19 per customer. From her thousands of records over the past year, she finds that her **cost per customer** (including food and labor) follows a normal distribution with a mean of $17.50 and a standard deviation of $3.00.

We'll help Jane make a few predictions. Assume that there is no inflation in the past year.

(a) What is the best point predictor of the cost for the next customer? (2 points)

(b) Jane wants to find a 90% prediction interval of customer costs, centered around the mean ($17.50). In other words, 90% of customers' costs should fall within this range. What is the width of that interval, i.e., the distance between the upper and lower bounds? (2 points)

(c) Suppose customers are i.i.d. On one day, Jane serves 100 customers, and on another, she serves 250 customers. Which day is more likely to have a higher average profit **per customer** (price minus cost per customer)? (2 points)

(d) Suppose customers are i.i.d. On one day, Jane serves 100 customers, and on another, she serves 250 customers. Which day is more likely to have an average profit **per customer** (price minus cost per customer) that is higher than $2.0 per customer? (2 points)

Question 3: Netflix Game Launch (18 points)

In this question, we will use the file `streaming_beta.csv` in the Bruin Learn folder for this homework. This file describes weekly household usage before and after the launch of the "beta" version of the Netflix app, where gaming features were introduced. Below is a brief data dictionary:

- `household_id`, `hh_size`, `hh_income`: household identifier, household size, and household income
- `has_netflix_pre`, `has_hulu_pre`, `has_disney_pre`: whether the household subscribed to Netflix, Hulu, and Disney+ *before* the beta launch
- `has_netflix_post`, `has_hulu_post`, `has_disney_post`: whether the household subscribed to Netflix, Hulu, and Disney+ *after* the beta launch

- `netflix_hours_pre`, `hulu_hours_pre`, `disney_hours_pre`: weekly hours spent on Netflix, Hulu, and Disney+ *before* the launch
- `netflix_hours_post`, `hulu_hours_post`, `disney_hours_post`: weekly hours spent on Netflix, Hulu, and Disney+ *after* the launch

The dataset includes 4,000 households. Such TV/app usage data can often be obtained from internet service providers. We will use this dataset to make business inferences and understand market patterns.

(a) Changes in Netflix Usage Among Subscribers

Netflix's goal is to increase weekly usage by at least 60 minutes per household among subscribers. In this part, use only households that subscribed to Netflix **both before and after** the beta launch, meaning `has_netflix_pre = 1` and `has_netflix_post = 1`. You are asked to determine whether the mean increase for this group exceeds 60 minutes per week. For this exercise, treat these households as an i.i.d. sample. To answer this question, you may need to subset the data first, For example:

```
sample <- data[data$has_netflix_pre == 1 & data$has_netflix_post == 1,]
```

(a-1) For each household in this group, calculate the change in Netflix usage by subtracting `netflix_hours_pre` from `netflix_hours_post`. Plot a histogram of these changes and briefly describe its shape. (2 points)

(a-2) Should you use a one-tailed or two-tailed hypothesis test? State the null and alternative hypotheses, and explain what the population mean in your hypotheses represents. Remember that the data are measured in hours. (2 points)

(a-3) A colleague suggests checking whether the usage changes in (a-1) follow a normal distribution before conducting the hypothesis test. Is it necessary for these individual changes to be normally distributed? Explain your answer. (2 points)

(a-4) Conduct the hypothesis test at a 5% significance level, using a 95% confidence level in R. Report the sample mean, test statistic, and p-value. Is there enough evidence that the mean increase for this group exceeds 60 minutes per week? (3 points)

(a-5) Does a statistically significant result in (a-4) show that the gaming features caused a greater-than-60-minute-per-week increase in Netflix usage? Explain. (2 points)

(b) Competition with Disney+ and Hulu

We want to investigate whether Netflix competes with Hulu and Disney+ for households' viewing time. Start with *all 4000 households* in `streaming_beta.csv`, including those excluded from part (a). In the data, each household contributes two observations: one before the beta launch and one after. Classify each observation as a Netflix subscriber or non-subscriber using its subscription status in that period.

(b-1) Using the data from both before and after the beta launch, calculate two averages of combined weekly Hulu and Disney+ usage: one for Netflix subscribers and one for non-subscribers. Which group has the higher average usage of Hulu and Disney+, the group with Netflix subscription or the group without? (2 points)

(b-2) Using the findings in (b-1), can you conclude that Netflix does not compete with Disney+ and Hulu or not? Explain. (1 point)

(b-3) Now let's focus on *switchers*: households that joined or canceled Netflix between the two periods. For each switcher, calculate its combined Hulu and Disney+ usage *with Netflix minus without Netflix*. State the null and alternative hypotheses, then use a one-sample t-test to test *whether the population mean of these differences is zero*, at a 5% significance level. Report the number of switchers, the mean difference, the p-value, and your conclusion. (4 points)
Hint: For households that joined Netflix, subtract before-period usage from after-period usage. For households that canceled Netflix, subtract after-period usage from before-period usage.

(b-4) Why might the comparison across all households in (b-1) and the comparison among only switchers in (b-3) give different results? Explain what the results suggest about Netflix's competition with Disney+ and Hulu. (2 bonus point)